

Simulation 2: Penney's Game

(Non-Transitive Probabilities)

Setup

Penney's Game is played by flipping a fair coin repeatedly until one of two pre-chosen length-3 patterns appears as a run of consecutive tosses. The player whose pattern shows up first wins. The simulation flips a fair coin (seeded for reproducibility), keeps a rolling window of the last three tosses, and stops the moment that window matches one of the two competing patterns. Each matchup was run for **100,000 races**.

Alongside the simulation, the code computes the **exact** odds using Conway's leading-number algorithm, so every empirical figure can be checked against theory.

Results

Matchup	Empirical P(first wins)	Exact P(first wins)	Exact odds	Winner
HTT vs HTH	0.5012	0.5000	4 : 4	tie (50/50)
HTH vs THH	0.4986	0.5000	3 : 3	tie (50/50)
THH vs TTH	0.3344	0.3333	2 : 4	TTH (2/3)

The simulation matches the exact theory to within ordinary Monte Carlo error (about ± 0.002 at 100,000 trials). Two of the three requested matchups are genuinely fair, while THH vs TTH is a decisive 2 : 1 win for TTH.

To expose the full structure, here is the exact probability that the row sequence beats the column sequence across all 8 patterns:

Why the races are not all 50/50

Each isolated 3-toss window does have probability $\frac{1}{8}$. But a race is **not** a contest of isolated windows. It is a question of which pattern the coin stream tends to *complete first* as tosses accumulate, and that depends entirely on how the two patterns **overlap** with each other and with themselves.

The mechanism is the shared prefix. Consider the two fair matchups:

- **HTT vs HTH.** Both begin with HT. Neither pattern can appear until an HT has occurred, so the race effectively starts only once HT shows up. After that HT, a single toss decides everything: a T completes HTT, an H completes HTH. One fair toss, so the split is exactly $\frac{1}{2}$ each. The shared two-symbol prefix erases any advantage.

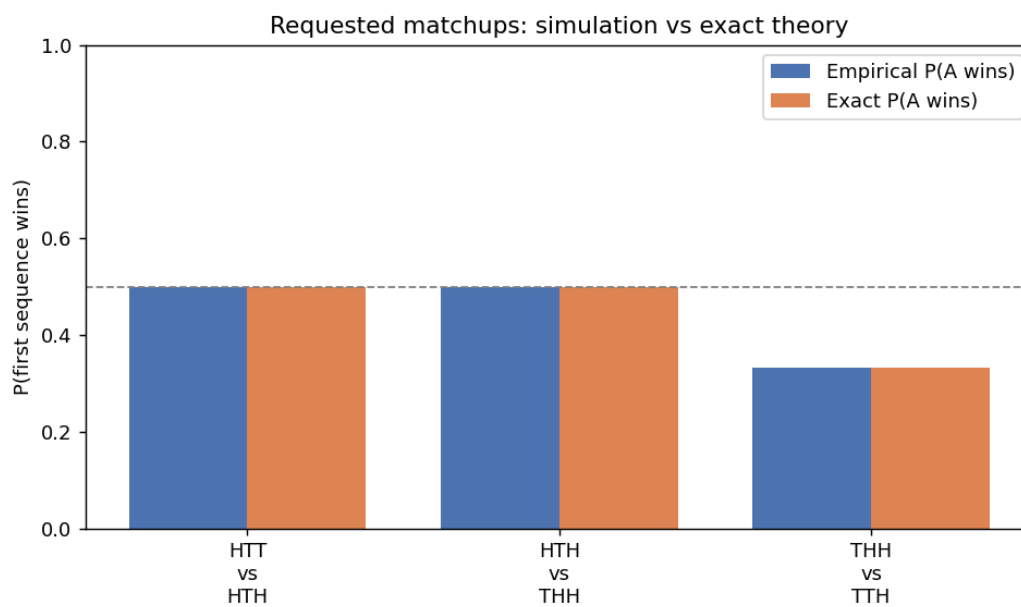


Figure 1: Requested matchups: empirical vs exact

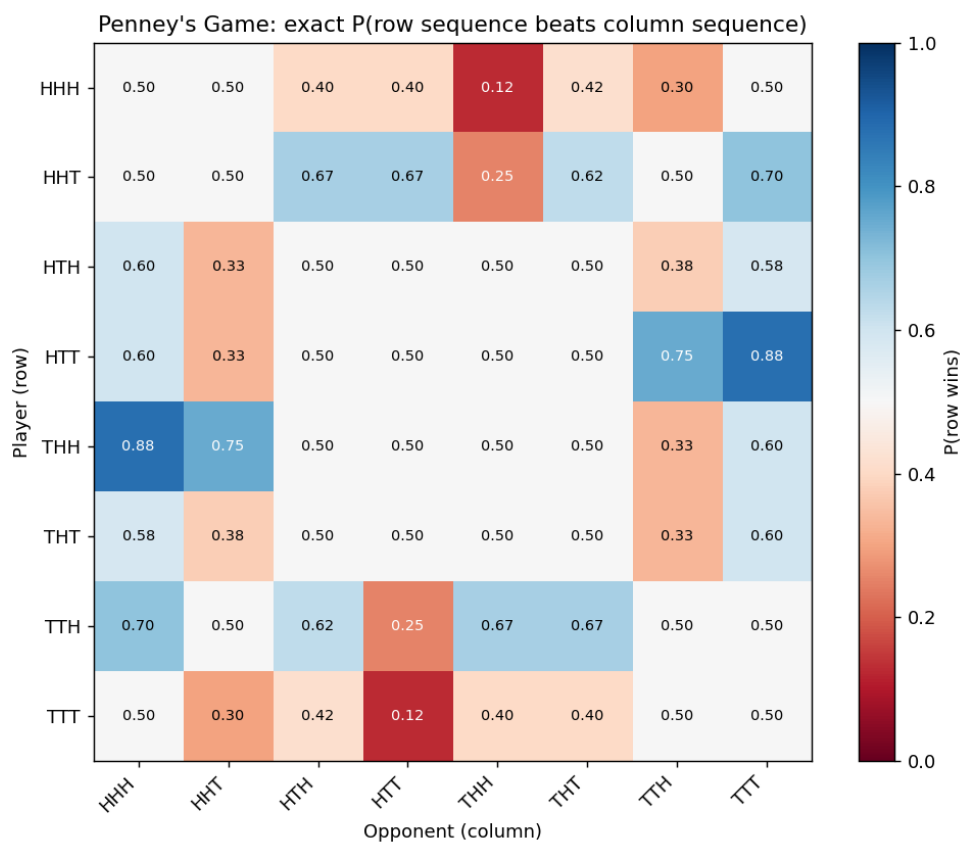


Figure 2: Full pairwise win matrix

- **HTH vs TTH.** By a symmetric overlap argument these also balance at $\frac{1}{2}$, which Conway's numbers confirm with odds 3 : 3.

Now the lopsided one:

- **THH vs TTH.** Here the patterns do **not** share the deciding structure in a symmetric way. The key fact is what happens after a T. Track the first time the stream produces TT versus TH. Once you see TT, you are one H away from TTH, and even if an H does not come immediately, the trailing T keeps you primed. THH, by contrast, needs the cleaner setup T then HH, and a stray T in the middle resets its progress back toward TTH's favor. The self-overlap of the patterns makes TTH reach its target first twice as often, giving the exact 2 : 1 result the simulation reproduces.

The general statement of this is **Conway's algorithm**. For patterns A and B define the correlation $C(X, Y)$ by scanning every shift $k = 0, \dots, L - 1$ and adding 2^{L-1-k} whenever the last $L - k$ symbols of X equal the first $L - k$ symbols of Y . This number measures how strongly the tail of X primes the head of Y . The odds that A beats B are then

$$\text{odds}(A : B) = (C(B, B) - C(B, A)) : (C(A, A) - C(A, B)).$$

Self-correlation $C(X, X)$ is large when a pattern overlaps itself (e.g. it can restart inside its own run), and that self-overlap is exactly what creates the asymmetry that single-window probabilities miss.

Why the winning relation is non-transitive

Because the winner depends on the *pairwise* overlap, there is no pattern that is best against all others. The "beats" relation contains cycles. One explicit loop, with exact probabilities from the simulation's Conway computation:

$$\text{THH} \xrightarrow{3/4} \text{HHT} \xrightarrow{2/3} \text{HTT} \xrightarrow{3/4} \text{TTH} \xrightarrow{2/3} \text{THH}.$$

Reading it out: THH beats HHT with probability $\frac{3}{4}$, HHT beats HTT with probability $\frac{2}{3}$, HTT beats TTH with probability $\frac{3}{4}$, and TTH beats THH with probability $\frac{2}{3}$, returning to the start. Each arrow is a strict majority, yet following the arrows brings you back where you began.

This is the same logic as rock-paper-scissors: dominance is a local, pairwise property, not a global ranking. The structural reason is that "beats" is built from overlap relationships between *specific pairs*, and overlap is not a transitive quantity. A pattern B chosen to exploit the tail structure of A can itself be exploited by a third pattern C targeting B 's tail, and C in turn is vulnerable to something looping back toward A . The heatmap above shows this directly: no row is uniformly blue (winning against everyone), and every strong column is beaten by some other row.

The practical upshot is the classic Penney swindle: in a two-player game where the first player must announce their length-3 pattern, the second player can always respond with a pattern that wins with probability at least $\frac{2}{3}$. Going second is the advantage, the opposite of most intuition about such games.

Reproducing the results

```
python penney_game.py
```

This prints the matchup table and the non-transitive cycle to the console, and writes `penney_heatmap.png`, `penney_matchups.png`, and `penney_results.txt`.